Programming Fundamentals

Lab 5 Submission

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Lab 5 – Task 1

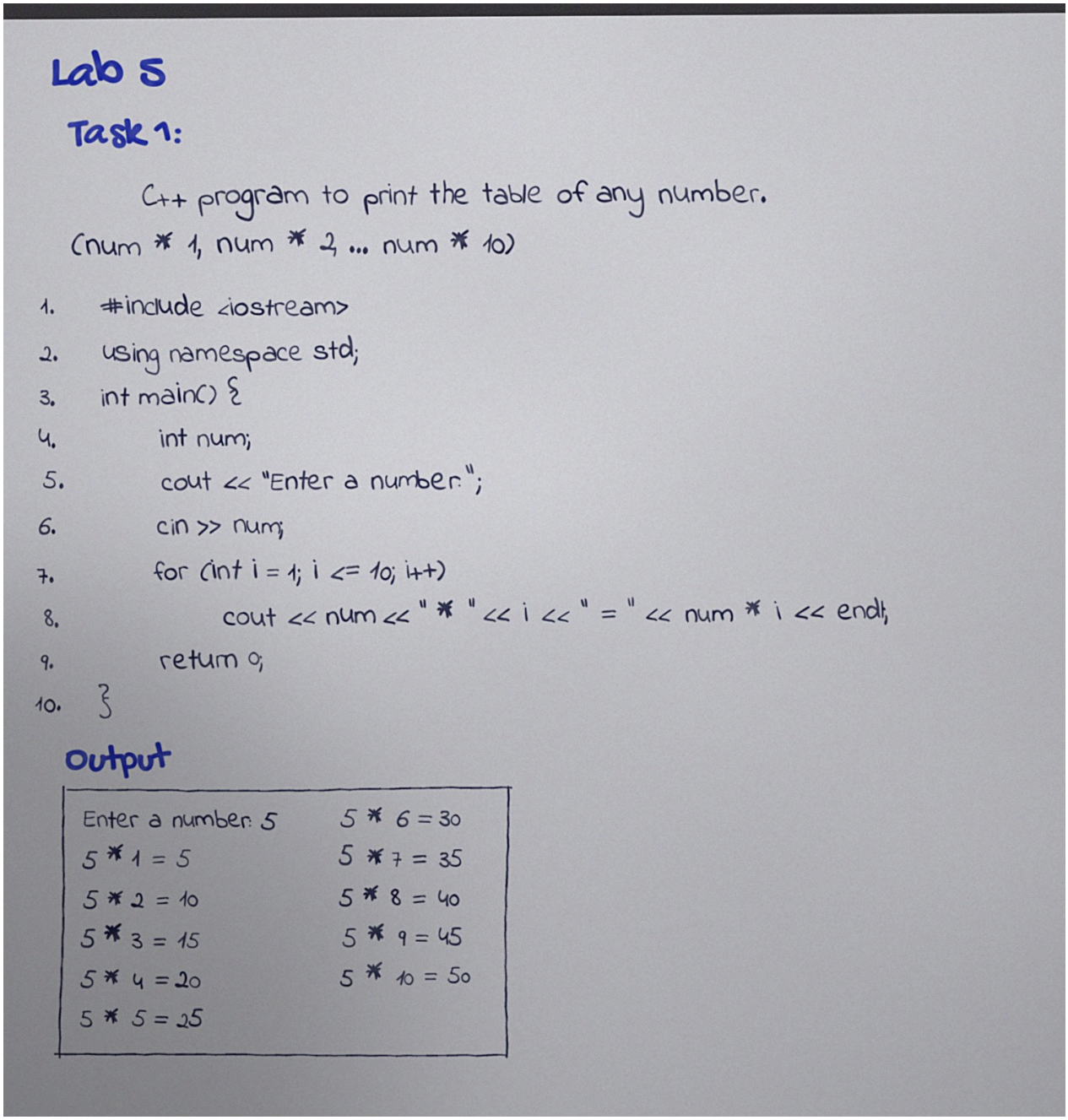
Question

C++ program to print the table of any number.

The concept of generating the table of any number is to multiply that number from 1 to 10:

```
num * 1, num * 2, num * 3, ... , num * 10
```

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int num;
    cout << "Enter a number: ";
    cin >> num;
    for (int i = 1; i <= 10; i++)
        cout << num << " * " << i << " = " << num * i << endl;
    return 0;
}
```

Output

```
Enter a number: 5
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```

Explanation

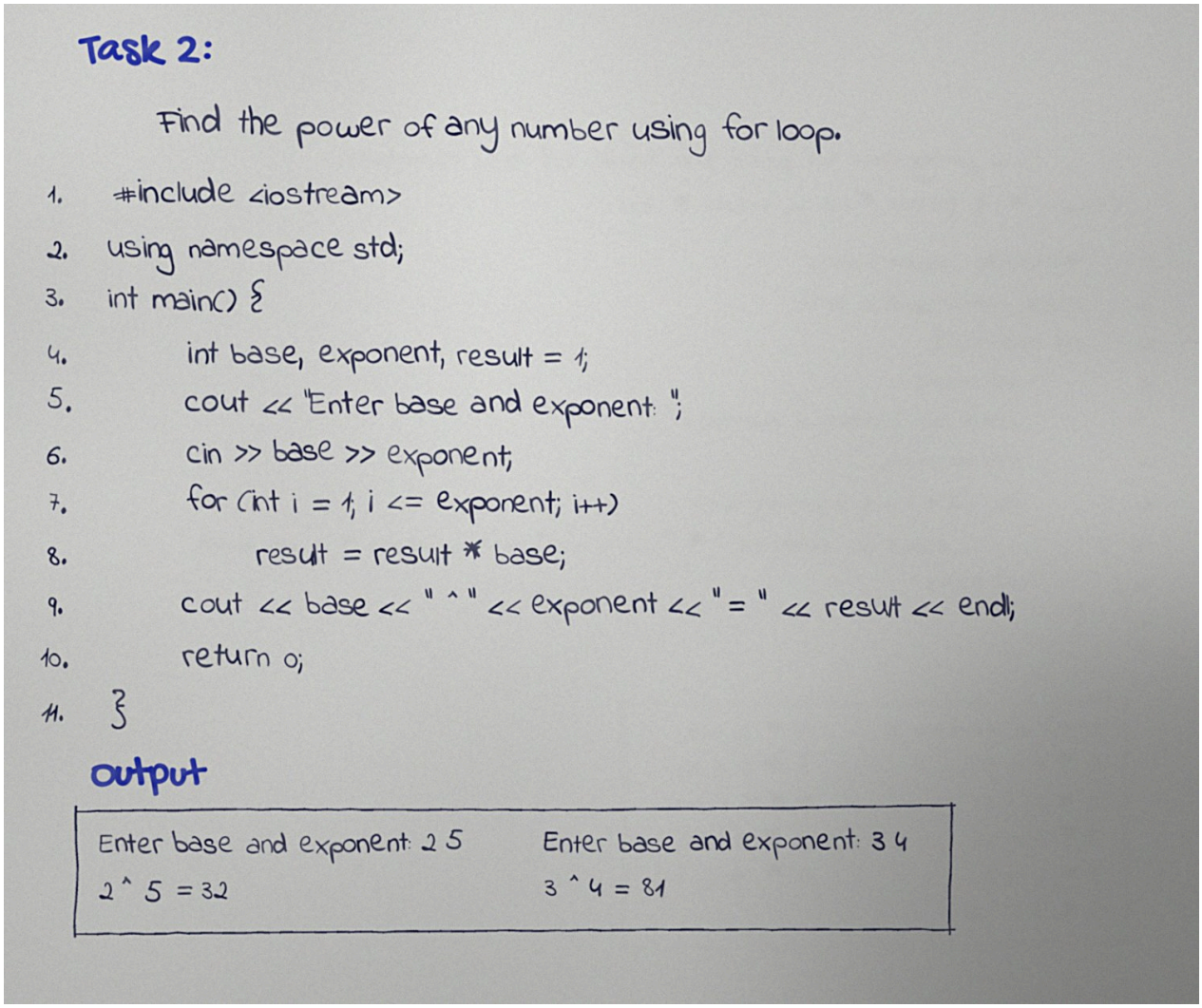
- The number is read into `num`, then the `for` loop runs with `i` going from 1 to 10.
- Each time the loop runs it prints one line of the table: `num * i`.
- `i++` adds 1 to `i` after every line, and the loop stops when `i` becomes 11.

Lab 5 – Task 2

Question

Find the power of any number using a for loop.

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int base, exponent, result = 1;
    cout << "Enter base and exponent: ";
    cin >> base >> exponent;
    for (int i = 1; i <= exponent; i++)
        result = result * base;
    cout << base << " ^ " << exponent << " = " << result << endl;
    return 0;
}
```

Output

```
Enter base and exponent: 2 5
2 ^ 5 = 32

Enter base and exponent: 3 4
3 ^ 4 = 81
```

Explanation

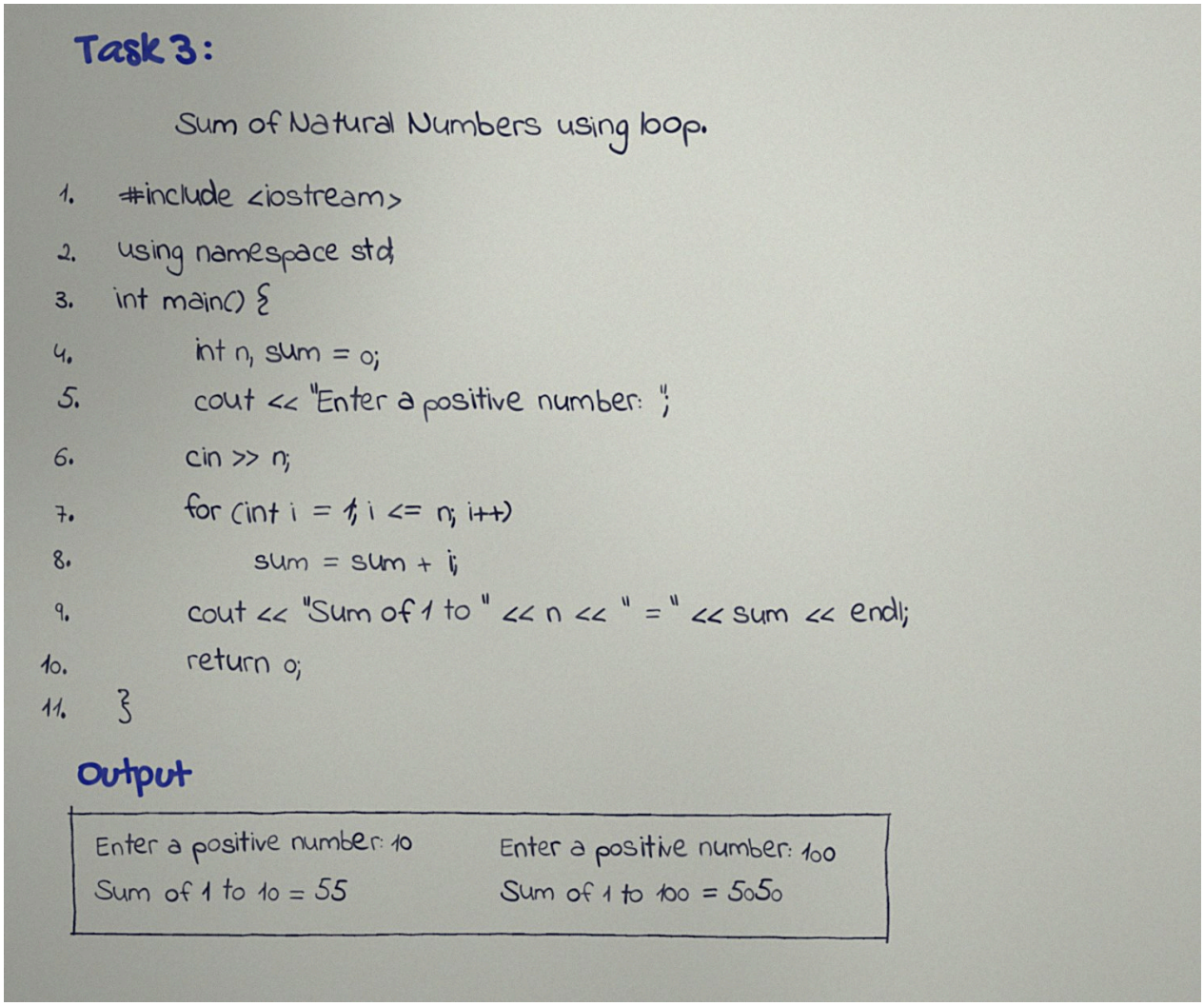
- `result` starts at 1, and the loop multiplies it by `base` once for every value of `i`.
- The loop runs `exponent` times, so 2 ^ 5 is 1 * 2 * 2 * 2 * 2 * 2 = 32.
- If the exponent is 0 the loop does not run at all, so the answer stays 1.

Lab 5 – Task 3

Question

Sum of natural numbers using a loop.

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int n, sum = 0;
    cout << "Enter a positive number: ";
    cin >> n;
    for (int i = 1; i <= n; i++)
        sum = sum + i;
    cout << "Sum of 1 to " << n << " = " << sum << endl;
    return 0;
}
```

Output

```
Enter a positive number: 10
Sum of 1 to 10 = 55

Enter a positive number: 100
Sum of 1 to 100 = 5050
```

Explanation

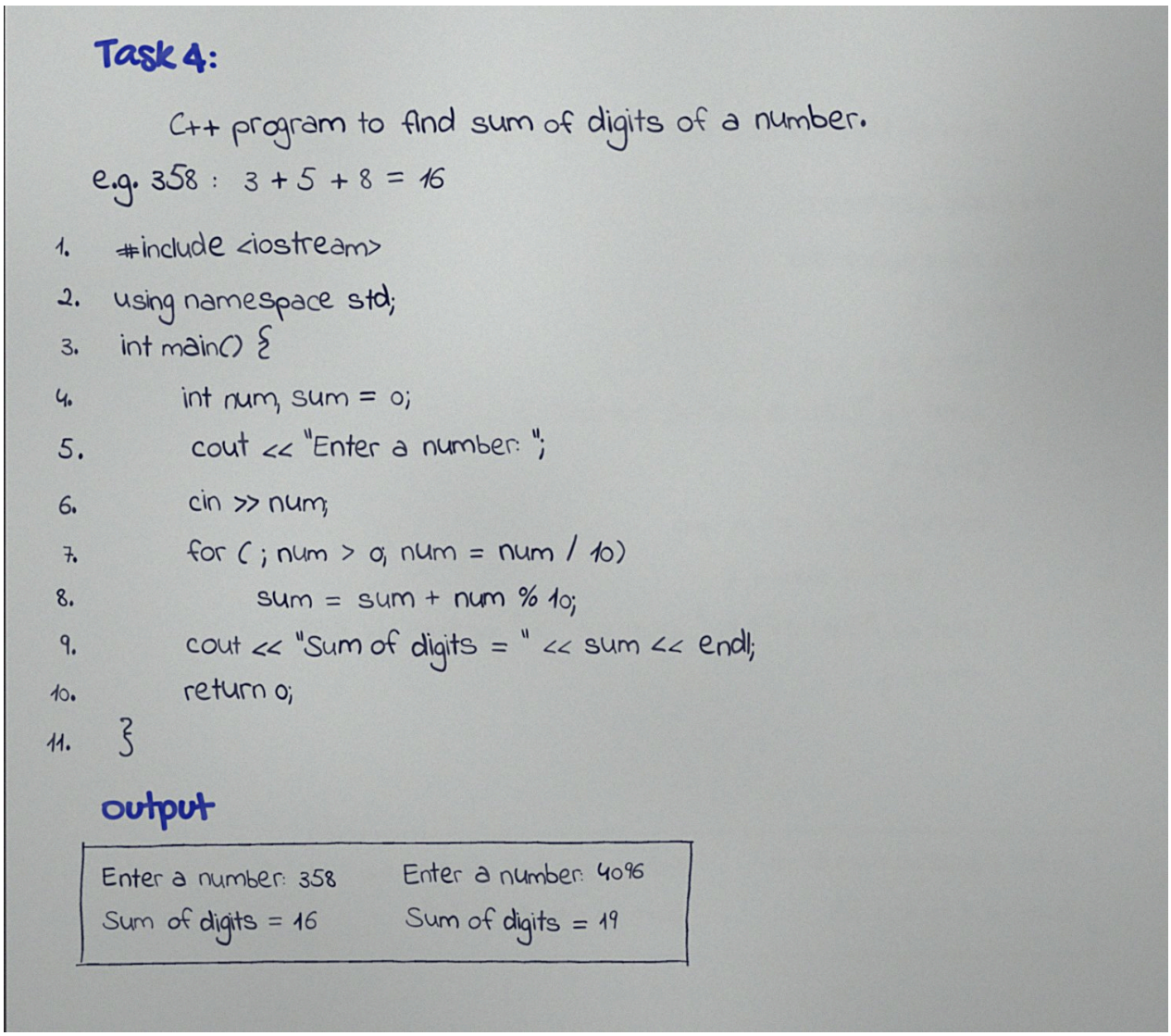
- Natural numbers are 1, 2, 3, ... so the loop starts `i` at 1 and goes up to `n`.
- Each value of `i` is added into `sum`, which starts at 0.
- For `n` = 10: 1 + 2 + 3 + ... + 10 = 55.

Lab 5 – Task 4

Question

C++ program to find the sum of digits of a number.
Sum of digits means adding all the digits of a number. For example, for 358 the sum of digits is 3 + 5 + 8 = 16.

Handwritten Code



Task 4:

C++ program to find sum of digits of a number.

e.g. 358 : 3 + 5 + 8 = 16

```
1.  #include <iostream>
2.  using namespace std;
3.  int main() {
4.      int num, sum = 0;
5.      cout << "Enter a number: ";
6.      cin >> num;
7.      for ( ; num > 0; num = num / 10)
8.          sum = sum + num % 10;
9.      cout << "Sum of digits = " << sum << endl;
10.     return 0;
11. }
```

output

Enter a number: 358	Enter a number: 4096
Sum of digits = 16	Sum of digits = 19

Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int num, sum = 0;
    cout << "Enter a number: ";
    cin >> num;
    for ( ; num > 0; num = num / 10)
        sum = sum + num % 10;
    cout << "Sum of digits = " << sum << endl;
    return 0;
}
```

Output

```
Enter a number: 358
Sum of digits = 16

Enter a number: 4096
Sum of digits = 19
```

Explanation

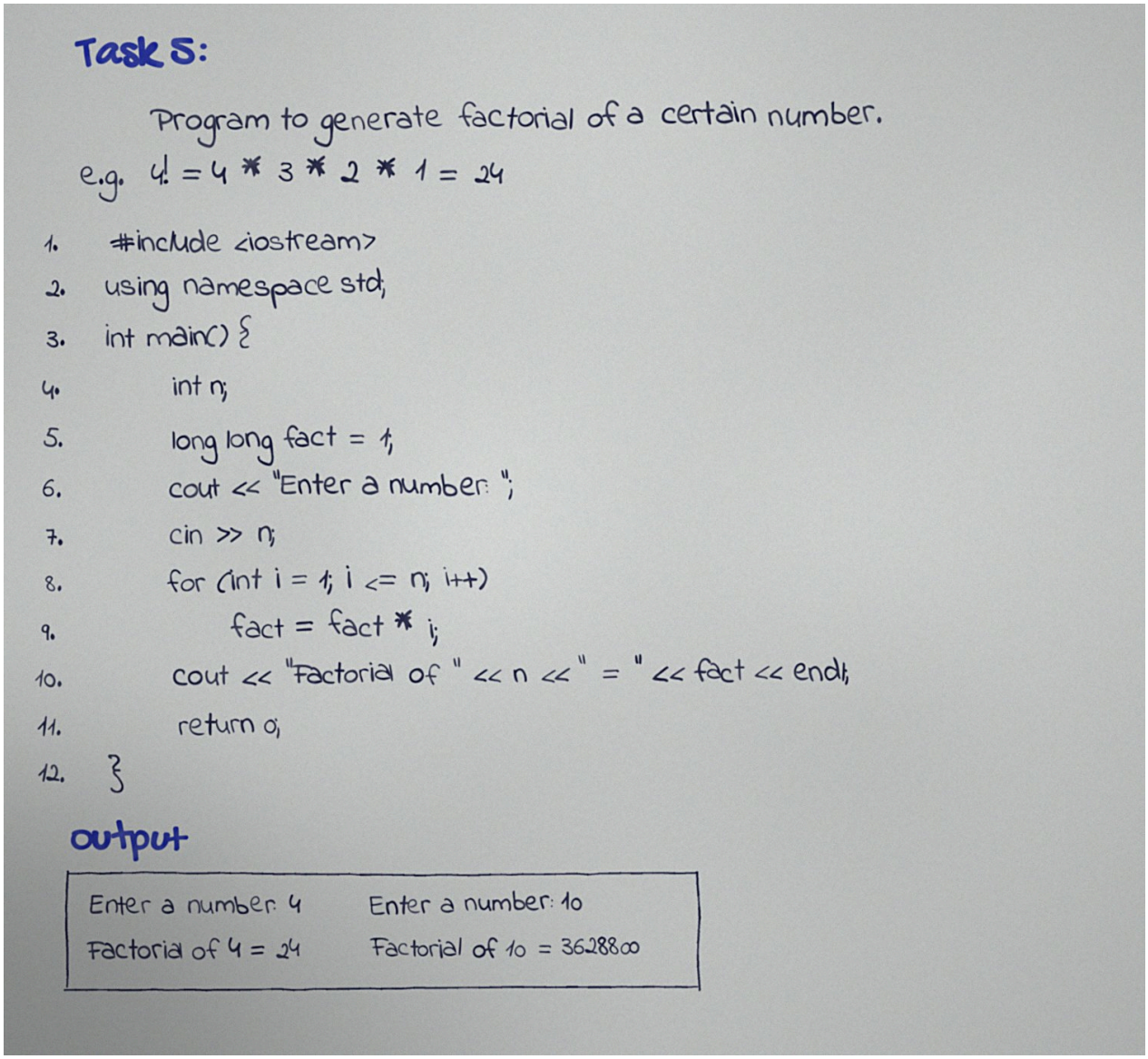
- `num % 10` gives the last digit, and `num / 10` removes the last digit.
- The initialization part of the `for` loop is left empty because `num` already has its value.
- For 358: 8 is added, then 5, then 3, and the loop stops when `num` becomes 0. The sum is 16.

Lab 5 – Task 5

Question

Program to generate the factorial of a certain number.
Factorial of any number is the product of that integer and all the integers below it, for example the factorial of 4 is 4! = 4 * 3 * 2 * 1 = 24.

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int n;
    long long fact = 1;
    cout << "Enter a number: ";
    cin >> n;
    for (int i = 1; i <= n; i++)
        fact = fact * i;
    cout << "Factorial of " << n << " = " << fact << endl;
    return 0;
}
```

Output

```
Enter a number: 4
Factorial of 4 = 24

Enter a number: 10
Factorial of 10 = 3628800
```

Explanation

- `fact` starts at 1 and the loop multiplies it by 1, 2, 3, ... up to `n`.
- For 4: 1 * 1 * 2 * 3 * 4 = 24.
- `long long` is used for `fact` because factorials grow very fast and can become too big for an `int`.

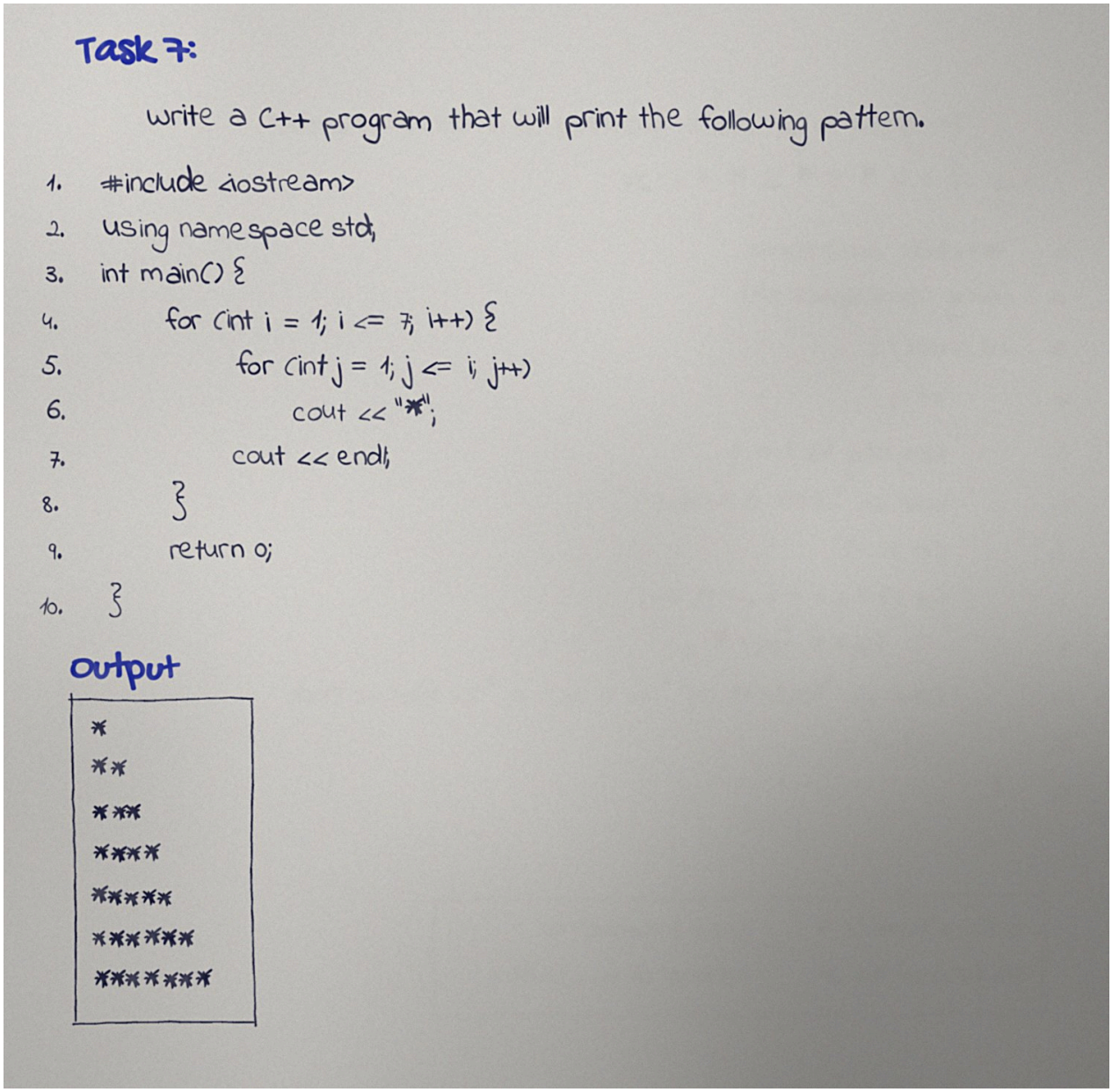
Lab 5 – Task 7

Question

Write a C++ program that will print the following pattern.

```
*
**
***
****
*****
******
*******
```

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 7; i++) {
        for (int j = 1; j <= i; j++)
            cout << "*";
        cout << endl;
    }
    return 0;
}
```

Output

```
*
**
***
****
*****
******
*******
```

Explanation

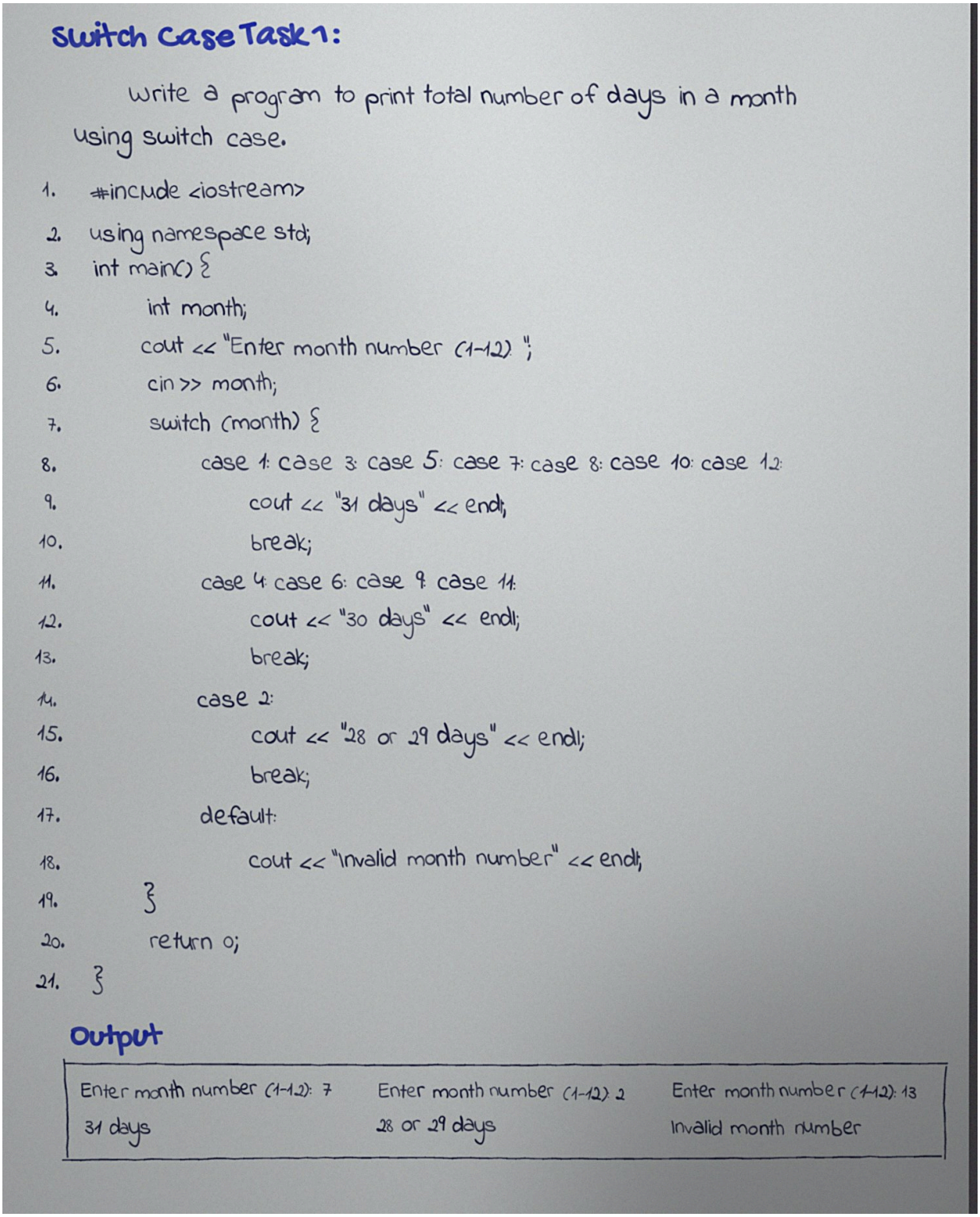
- This uses a nested loop. The outer loop (`i`) is the row number, from 1 to 7.
- The inner loop (`j`) prints `i` stars in that row, so row 1 has 1 star, row 2 has 2 stars, and so on.
- `cout << endl;` after the inner loop moves to the next line before the next row starts.

Lab 5 – Switch Case Task 1

Question

Write a C program to print the total number of days in a month using switch case.

Handwritten Code



Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    int month;
    cout << "Enter month number (1-12): ";
    cin >> month;
    switch (month) {
        case 1: case 3: case 5: case 7: case 8: case 10: case 12:
            cout << "31 days" << endl;
            break;
        case 4: case 6: case 9: case 11:
            cout << "30 days" << endl;
            break;
        case 2:
            cout << "28 or 29 days" << endl;
            break;
        default:
            cout << "Invalid month number" << endl;
    }
    return 0;
}
```

Output

```
Enter month number (1-12): 7
31 days

Enter month number (1-12): 2
28 or 29 days

Enter month number (1-12): 13
Invalid month number
```

Explanation

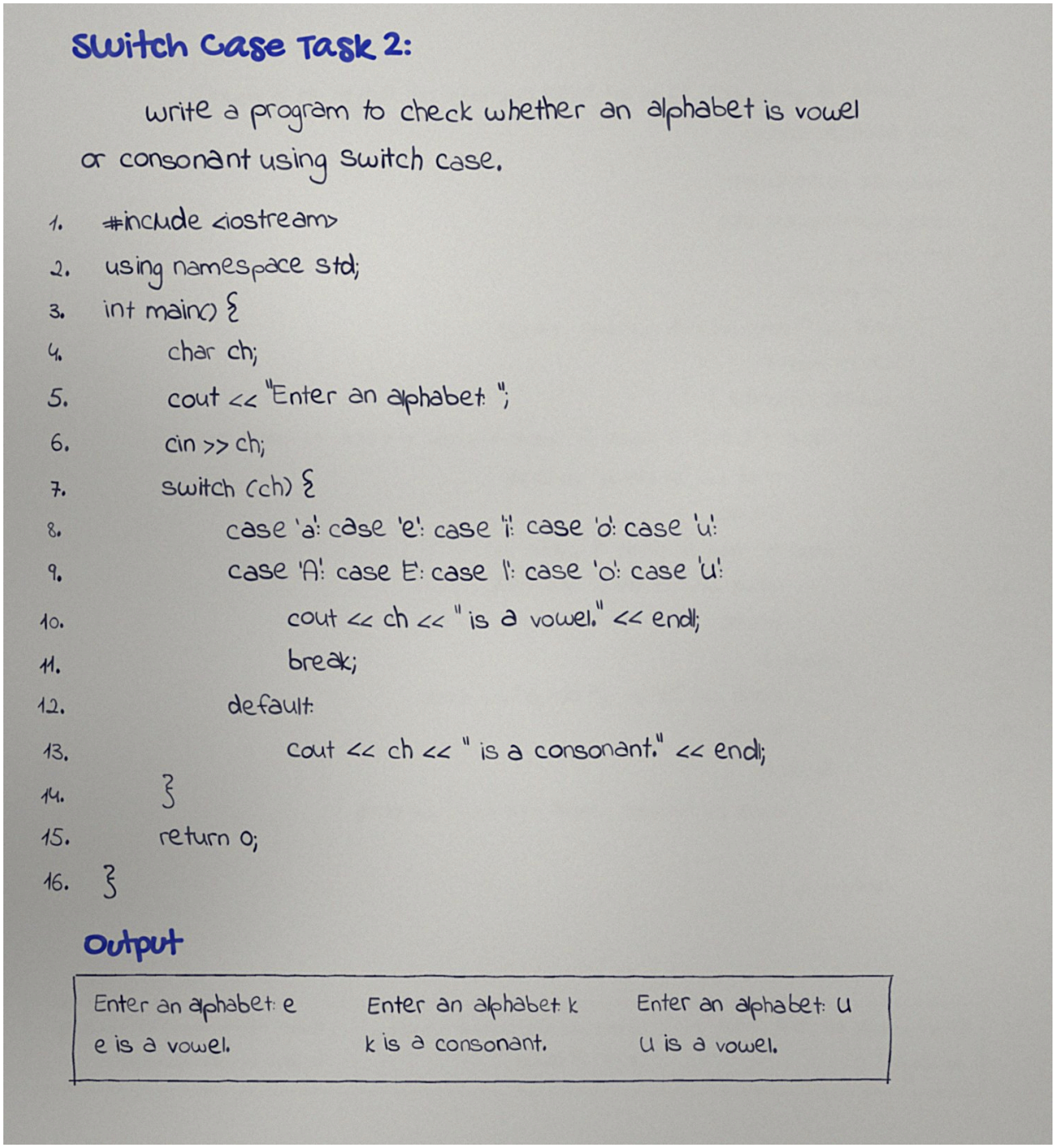
- `switch (month)` jumps to the `case` that matches the month number.
- Months with the same number of days share one block, e.g. `case 4: case 6: case 9: case 11:` all print 30 days.
- `break;` stops the switch after printing, and `default` handles any number that is not 1 to 12. February shows 28 or 29 days because it has 29 days in a leap year.

Lab 5 – Switch Case Task 2

Question

Write a C program to check whether an alphabet is a vowel or consonant using switch case.

Handwritten Code



Switch Case Task 2:

write a program to check whether an alphabet is vowel or consonant using switch case.

```
1.  #include <iostream>
2.  using namespace std;
3.  int main() {
4.      char ch;
5.      cout << "Enter an alphabet: ";
6.      cin >> ch;
7.      switch (ch) {
8.          case 'a': case 'e': case 'i': case 'o': case 'u':
9.          case 'A': case E: case I: case 'o': case 'U':
10.             cout << ch << " is a vowel." << endl;
11.             break;
12.         default:
13.             cout << ch << " is a consonant." << endl;
14.     }
15.     return 0;
16. }
```

Output

Enter an alphabet: e e is a vowel.	Enter an alphabet: k k is a consonant.	Enter an alphabet: U U is a vowel.

Code Written on Phone / Desktop

```cpp
#include <iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter an alphabet: ";
    cin >> ch;
    switch (ch) {
        case 'a': case 'e': case 'i': case 'o': case 'u':
        case 'A': case 'E': case 'I': case 'O': case 'U':
            cout << ch << " is a vowel." << endl;
            break;
        default:
            cout << ch << " is a consonant." << endl;
    }
    return 0;
}
```

Output

```
Enter an alphabet: e
e is a vowel.

Enter an alphabet: k
k is a consonant.

Enter an alphabet: U
U is a vowel.
```

Explanation

- `switch` works with characters, so each vowel is written as a `case` with single quotes, e.g. `'a'`.
- Both small and capital vowels are listed, so `A` and `a` are both vowels.
- If the letter is not one of the 10 vowel cases, the `default` part runs and prints consonant.

Lab 5 – Think??

Question

What will be the output of the following programs?

Q1. What will be the output of the C program?

```
#include<stdio.h>
int main()
{
    for(5;2;2)
    {
        cout<<"Hello";
    }
    return 0;
}
```

Q2. What will be the output of the C program, if input is 6?

```
#include<stdio.h>
int main()
{
    int i;
    for(i = 0; i>9; i+=3)
    {
        cout<<"for ";
    }
    return 0;
}
```

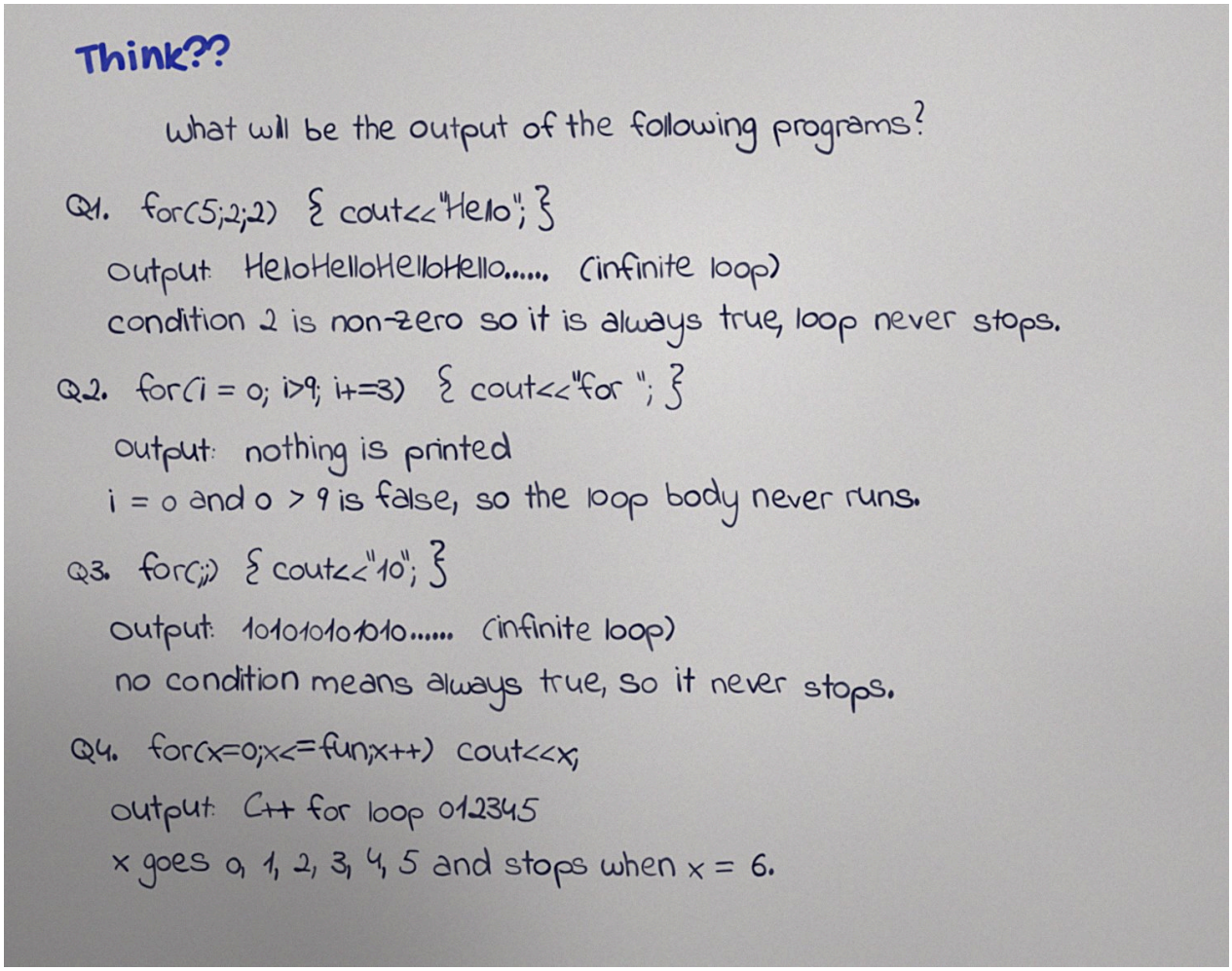
Q3. What will be the output of the C++ program?

```
#include<stdio.h>
int main()
{
    for(;;)
    {
        cout<<"10";
    }
    return 0;
}
```

Q4. What will be the output of the C program?

```
int main()
{
    int fun=5;
    cout<<"C++ for loop ";
    int x = 5;
    for(x=0;x<=fun;x++)
    {
        cout<<x;
    }
    return 0;
}
```

Handwritten Answers



Output (Q1)

```
HelloHelloHelloHelloHelloHello...
(prints Hello again and again, it never stops)
```

Output (Q2)

```
(nothing is printed)
```

Output (Q3)

```
1010101010101010101010...
(prints 10 again and again, it never stops)
```

Output (Q4)

```
C++ for loop 012345
```

Explanation

- Q1: `5` and the update `2` do nothing. The condition `2` is not zero, so it is always true and the loop never ends. It is an infinite loop that keeps printing Hello.
- Q2: `i` starts at 0 and the condition `i>9` is false at the very start, so the body never runs and nothing is printed. The program never reads any input, so entering 6 changes nothing.
- Q3: `for(;;)` has no condition, which C++ treats as always true, so it is an infinite loop printing 10.
- Q4: "C++ for loop " is printed first. Then `x` is set to 0 and the loop prints 0, 1, 2, 3, 4, 5 with no spaces, and stops when `x` becomes 6.
- Note: these programs use `cout`, which needs `#include <iostream>` and `using namespace std;`. With only `<stdio.h>` the compiler gives an error, so the answers assume those two lines are added.

END OF ASSIGNMENT

```
return 0;   // program finished successfully
```

Thank you for reading!